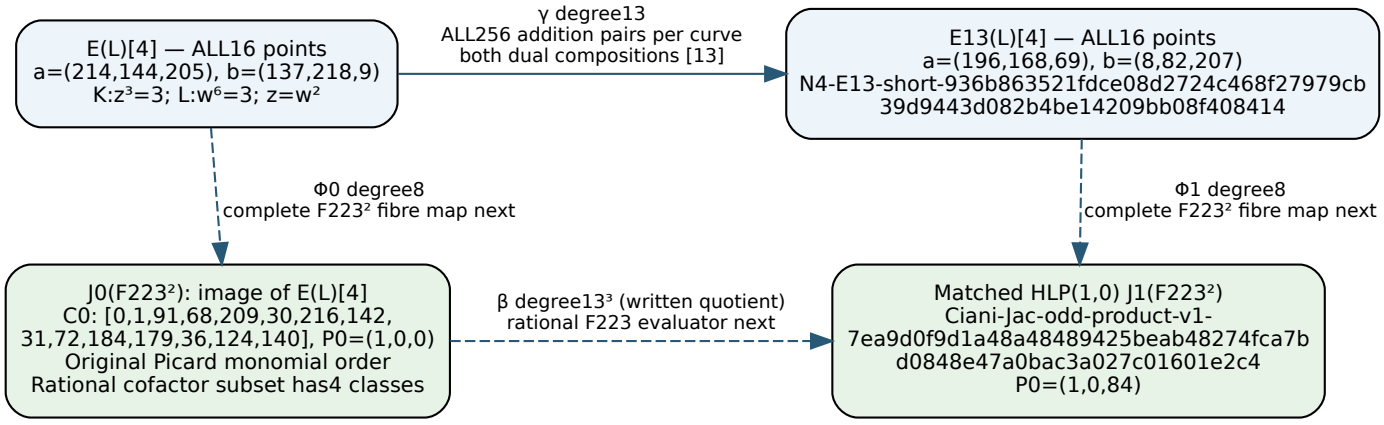
Quadratic lifts for the matched Jacobian quotient

2026-11-14 · fb21de / 7f5766 / b28edc · unsigned / finite and conditional scope



The checked conditional lemma constructs a quadratic lift: if Phi is a surjective abelian-group homomorphism commuting with Frobenius, ker(Phi) is killed by2 and F³ fixes that kernel, every F-fixed x has an F²-fixed lift. Choose a geometric lift a, put c=F(a)−a, and take b=a+F(c). Then Phi(b)=x and F²(b)=a+c+F(c)+c=b. All assumptions remain explicit.

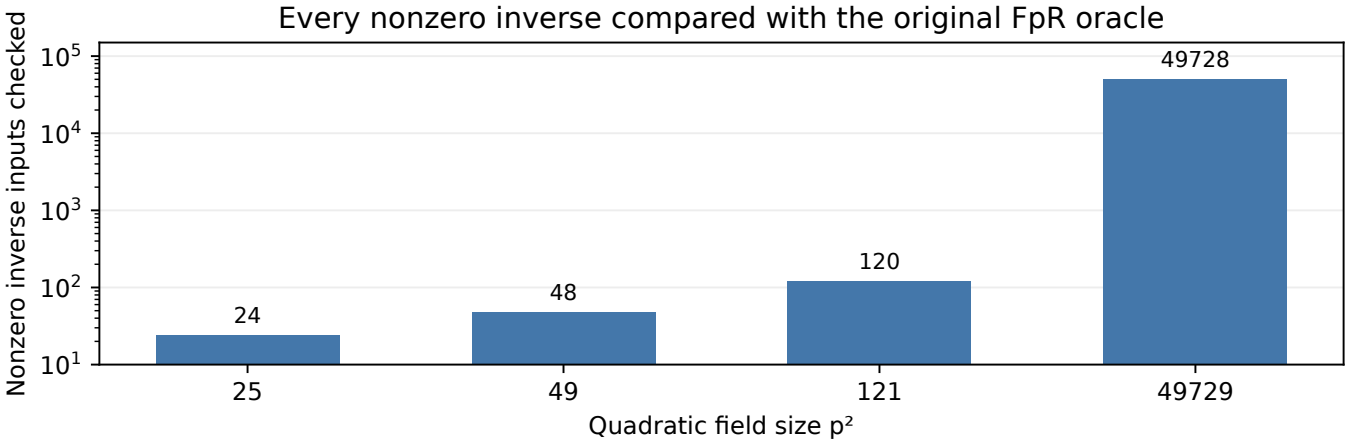
For the written Ciani quotient A=Res(K/F223)E→J0, the actual split E[2]/K gives F³=1 on A[2]. A cofactor-four target has 2N(x)=0, so N Phi=[2] forces every lift to have order dividing4. The remaining cofactor calculation therefore needs at most sixteen actual E[4](L) inputs.

Nativefb21de retains all32 source/target four-torsion point records, including both identities, every256 ordered addition pairs per curve, both degree13 compositions, and the actual Frobenius action. In A(F223²)=E(L), base Frobenius is coordinate F³; coordinate F has actual off-curve mutation examples. The coefficient subfield is v=w³, v²=3.

The finite C4×C2 control checks all group, homomorphism, automorphism and kernel-two conditions but drops kernel F³=1. Its nonzero target has no F²-fixed lift. Lean checks this finite control by kernel decide; the six named structural roots use no axioms. Both original gates pass20 declarations including seven named roots and generated helpers, zero trusted. Actual geometry remains a written/native bridge.

Sources: Milne AV(2008), I§7 and Remark8.12 printed39; HLP math/9809210v1 §4.1/Lemma13/Proposition15, including twist assumptions. Primary review622871; prospective5be5e0 was published in3422ccb4 before code. All claims remain unsigned. Dashed edges show the native map work still to implement.

Genuine quadratic Picard arithmetic: checked coverage



p / ν	Field size	Inverse inputs	Base-class control	Addition edges
5 / 2	25	24	148 complete	888
7 / 3	49	48	356 complete	2848
11 / 2	121	120	1324 complete	15888
223 / 3	49729	49728	17 ordered inputs	289

Nativeb28edc passes two unfiltered tests, zero failed/ignored/filtered. Each nonzero inverse and every field Frobenius encoding agrees with the original FpR. The small-field base closures have148,356,1324 distinct classes, and all generating edges agree across original prime Picard, copied prime code and the genuine quadratic backend. The223 case checks17 ordered base inputs/289 pairs; it does not enumerate the entire Jacobian.

For p=5,7,11, complete extension-curve point counts match an independent original-FpR calculation. Genuine non-base-field classes pass64 Frobenius/commutativity pairs and512 associativity triples per field. Constructors reject square/zero/noncanonical ν and composite p² as a prime field. Encoded ν=p squares toν; the incorrect integer ring modulo p² squares it to0.

The original Picard file and its tests are unchanged. The isolated production derivative inventories17 literal edits, including direct modular shortcuts in evaluation, gradient and tangent setup. New-source, original784 and arithmetic22 guards return0. Native/proof failures and caller/workspace/path errors remain preserved; none is relabeled as a mathematical obstruction.

Quadratic norm multiplicativity has exact Int Lean binding637454, original native certificate IDC1hd724979553e97e07, exact integer coefficient expansion,64 sealed plus32 extra checker points, and an actual missing-term process rejection101. The certificate pointer names the checked source; the native checker itself does not run Lean. Original norm gates pass one declaration/zero trusted. All standings remain unsigned.

Next: extend the actual complete line/conic fibre ideals and origin over F223², trace all three F² conjugates, match every rational cofactor class, then evaluate both canonical beta directions. Odd-subgroup halving alone is insufficient. Complete global closed rational coordinates and calibrated costs remain separate requirements. No timing or advantage is claimed.

Exact identity: K=F223[z]/(z³−3), ell2773073, #E(K)=11092292. Matched target coefficients: [1,121,174,193,172,175,11,124,77,78,48,58,89,63,164], P0=(1,0,84). Full models, coordinates and source links are in BETA_RATIONAL_LIFT_DERIVATION_01 and the native evidence.